Matthew (Moslem) Shokrolahi

Toronto, ON | moslem.sh.99@gmail.com | (613) 449-7401 | LinkedIn | GitHub | Google Scholar

SUMMARY

Applied ML Scientist (Ph.D., Computer Engineering) with production experience building and evaluating LLM, VLM and generative-image systems. Built LLM-enabled RAG for e-commerce recommendations (+23% MRR), fine-tuned multimodal models, and developed production ML evaluation pipelines at a major bank. Published at ICML and TMLR on controllable text-to-image generation. Strong in Python, MLOps (MLflow, Docker, CI/CD) and communicating technical results to non-technical stakeholders.

CORE EXPERTISE

LLM Applications & Fine-Tuning | RAG & Retrieval Systems | Deep Learning | Generative AI | Diffusion Models | MLOps | AWS | Databricks | Continual Learning | Responsible AI / Fairness & Bias

PROFESSIONAL EXPERIENCE

Applied Machine Learning Scientist: Model Risk Management & Compliance Oversight | TD Bank | Toronto, ON **| Jan 2026 - Present**

- Research and apply machine-learning and deep-learning techniques to large-scale banking datasets, evaluating predictive performance, stability, explainability, and model behavior.
- Develop production-ready ML evaluation pipelines for fraud, insurance, and marketing models, translating research methodologies into scalable tooling in collaboration with engineering and model-development teams.
- Research evaluation methodologies for LLM, VLM, and agentic AI systems, including bias/fairness, explainability, robustness, and model-behavior assessment.
- Validate the Credit Limit Increment model on Databricks and AWS SageMaker, assessing predictive performance, stability, explainability, and fairness at scale.

AI Consultant | KnowQuest | Toronto, ON **| Sep 2023 - Sep 2025**

- Developed a CLIP-based approach to detect AI-generated or mismatched review images using image-text alignment, improving fake review detection precision by 18%.
- Enhanced the feedback and recommendation system with an LLM-enabled RAG framework, improving mean reciprocal rank (MRR) by 23% and strengthening recommendation relevance.

Research Assistant | UAI Lab, Queen's University | Kingston, ON **| May 2021 - Jul 2026**

- Developed interpretable and controllable methods for safe and fair text-to-image diffusion, improving demographic fairness while maintaining image quality and background consistency.
- Analyzed the architectures of U-Net and transformer-based diffusion foundation models (SDXL, PixArt-α, FLUX.1, SD3.5), using attention-head interpretability to identify the components that encode semantic concepts and to enable linear, controllable concept steering for responsible generation. Analyzed the architectures of U-Net and transformer-based diffusion foundation models, using attention-head interpretability to identify the components that encode semantic concepts and to enable linear, controllable concept steering for fair and safe generation.
- Developed continual learning methods using deep metric learning to reduce catastrophic forgetting and inter-task confusion; also investigated zero-shot machine unlearning, training models on GPU-accelerated HPC clusters (SLURM).

Machine Vision Engineer | SAESPADAN | Isfahan **| Sep 2018 - May 2021**

- Designed a real-time stone-classification vision system using an ensemble of ResNet and EfficientNet models, reducing human error by 40% and increasing throughput by about 30% through weighted model averaging

TECHNICAL SKILLS

Software Systems & ML Engineering: AWS, GPU-accelerated training, SLURM/HPC, Docker, FastAPI, REST APIs, MLflow, FAISS/vector search, Git/GitHub, GitHub Actions/CI/CD, Linux, Jupyter, LangChain, LangGraph

Programming: Python (expert), SQL, C++ (proficient)

Frameworks & Libraries: PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn, NumPy, Pandas, OpenCV

EDUCATION

Ph.D. in Computer Engineering | **Queen's University, Kingston, ON** | May 2021 - Jul 2026

Dissertation: Adaptive and Responsible Machine Learning: From Incremental Learning to Unlearning and Safe Generation

M.Sc. in Computer Engineering | **Isfahan University of Technology (IUT), Isfahan** | Sep 2015

INDEPENDENT PROJECT AND PROFESSIONAL DEVELOPMENT

Smart Resume Screening: Developing a resume-screening workflow that ingests PDF/DOCX resumes and technical job descriptions, extracts structured skills, experience, education, and certifications, and separates identity information from ranking features.

Agentic AI Evaluation 1.0 Bootcamp | **Vector Institute** | Feb - Apr 2026

SELECTED PUBLICATIONS

- Shokrolahi, Sayedmoslem, Jae-Mo Kang, and Il-Min Kim. “Responsible Text-to-Image Diffusion: Interpretable and Linearly Controllable Semantics for Fair and Safe Generation.” Forty-third International Conference on Machine Learning (ICML), 2026.
- Shokrolahi, Sayedmoslem, Jae-Mo Kang, and Il-Min Kim. “Learning Where It Matters: Responsible and Interpretable Text-to-Image Generation with Background Consistency.” Transactions on Machine Learning Research (TMLR), 2026.
- Shokrolahi, Sayedmoslem, and Il-Min Kim. “MaxDiv: Zero-Shot Machine Unlearning via Distributionally Divergent Erasing Samples.” IEEE Transactions on Artificial Intelligence (TAI), 2025.
- Shokrolahi, Sayedmoslem, and Il-Min Kim. “Combating Inter-Task Confusion and Catastrophic Forgetting by Metric Learning and Re-Using a Past Trained Model.” Transactions on Machine Learning Research (TMLR), 2025.